

Equations for Étendue

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November 6, 2025

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1 Symbols

The units are a bit of a mess here; to make the equations a bit more transparent we work in photons, seconds, square meters, and steradians, so conversions from magnitudes will be necessary in places given common practice.

- α is the brightness of the target being sought in units commensurate with photons/second/meter².
- β is the brightness of the sky (photons/second/meter²/steradian).
- γ will be the SNR ratio.
- Ω is the area of the FOV of the sensor/optical system (steradians²).
- ω is the area of the FOV of a pixel (steradians²) or the seeing disk or the diffraction disk, whichever is largest. Basically this is the area over which the light is spread and needs to compete with the background.
- A is the effective collecting area (meter²).
- T is the integration time (seconds.) This may be the result of processing n shorter integrations t .
- g is used in some places as the limiting flux in photons/second/meter².
- η is the net quantum efficiency of the whole system
- f is the fraction of light from an object captured in the photometry aperture
- C is the calibration which gives the number of photons per second per square meter for a 0 magnitude object for the band being used.

2 Derivation of Search Rate

The signal from an object is $S \propto \alpha A T \eta f$. Assuming we are background limited, the noise is $N \propto \sqrt{\beta A T \eta \omega}$. Putting things together

$$SNR^2 = \gamma^2 = \frac{\alpha^2 A^2 T^2 \eta^2 f^2}{\beta A T \eta \omega} \quad (1)$$

$$\gamma^2 = \frac{\alpha^2 A T \eta f^2}{\beta \omega} \quad (2)$$

so

$$\boxed{T = \frac{\gamma^2 \beta \omega}{\alpha^2 A \eta f^2}} \quad (3)$$

This the “dwell time” required to reach the given SNR for the given object (assuming BLIP).

To get the search rate, multiply by the total area surveyed:

$$\boxed{SR = \frac{\Omega}{T} = \alpha \left(\frac{\alpha}{\gamma} \right)^2 \frac{A \Omega \eta f^2}{\omega \beta}} \quad (4)$$

which has units commensurate with steradians/second.

3 Conventional Units: Magnitudes

Typically, we will be numbers in magnitudes in a selected band; this will be magnitude for a source or magnitude per square arc second for a background number. For each band there is a calibration C_b which gives the number of photons per second per square meter for the filter band b in question so the flux f is given in terms of the magnitude m

$$m = -2.5 \log_{10}(f/C_b) \tag{5}$$

and

$$f = C_b 10^{-0.4m} \tag{6}$$